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The Universe and the Bucket

IS SPACE A HUMAN ABSTRACTION OR A PHYSICAL ENTITY?

It’s not often that a bucket of water is the central character in a three-hundred-year-long debate. But a bucket that belonged to Sir Isaac Newton is no ordinary bucket, and a little experiment he described in 1689 has deeply influenced some of the world’s greatest physicists ever since. The experiment is this: Take a bucket filled with water, hang it by a rope, twist the rope tightly so that it’s ready to unwind, and let it go. At first, the bucket starts to spin but the water inside remains fairly stationary; the surface of the stationary water stays nice and flat. As the bucket picks up speed, little by little its motion is communicated to the water by friction, and the water starts to spin too. As it does, the water’s surface takes on a concave shape, higher at the rim and lower in the center, as in Figure 2.1.

That’s the experiment—not quite something that gets the heart racing. But a little thought will show that this bucket of spinning water is extremely puzzling. And coming to grips with it, as we have not yet done in over three centuries, ranks among the most important steps toward grasping the structure of the universe. Understanding why will take some background, but it is well worth the effort.

